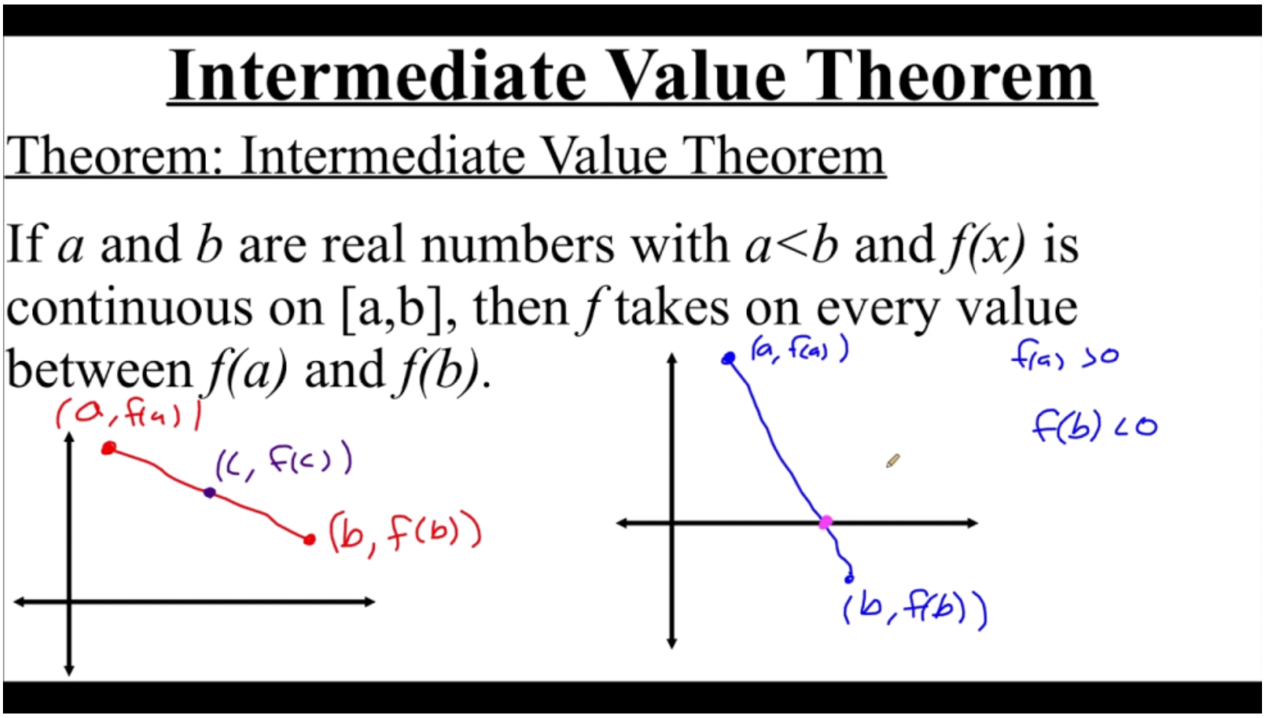
Exercise 03

Friday, 4 September 2026 4:19 pm



4. Use the Bisection method to find solutions accurate to within 10^{-2} for $x^4 - 2x^3 - 4x^2 + 4x + 4 = 0$ on each interval.

a. $[-2, -1]$ **b.** $[0, 2]$ **c.** $[2, 3]$ **d.** $[-1, 0]$

a.

n	a	b	$c = (a+b)/2$	$f(c)$
1	-2.000000	-1.000000	-1.500000	0.812500
2	-1.500000	-1.000000	-1.250000	-0.902344
3	-1.500000	-1.250000	-1.375000	-0.288818
4	-1.500000	-1.375000	-1.437500	0.195328
5	-1.437500	-1.375000	-1.406250	-0.062667
6	-1.437500	-1.406250	-1.421875	0.062263

b.

n	a	b	c	$f(c)$
1	0.000000	2.000000	1.000000	3.000000
2	1.000000	2.000000	1.500000	-0.687500
3	1.000000	1.500000	1.250000	1.285156
4	1.250000	1.500000	1.375000	0.312744
5	1.375000	1.500000	1.437500	-0.186508
6	1.375000	1.437500	1.406250	0.063676
7	1.406250	1.437500	1.421875	-0.061318

c.

n	a	b	c	$f(c)$
1	2.000000	3.000000	2.500000	-3.187500
2	2.500000	3.000000	2.750000	0.347656
3	2.500000	2.750000	2.625000	-1.757568
4	2.625000	2.750000	2.687500	-0.795639
5	2.687500	2.750000	2.718750	-0.247467
6	2.718750	2.750000	2.734375	0.044125

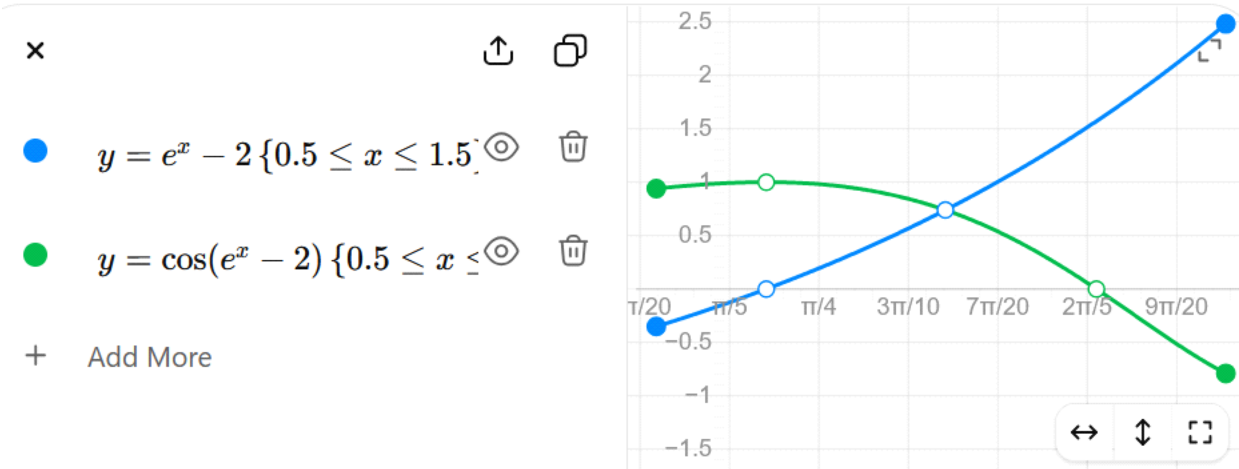
d.

n	a	b	c	$f(c)$
1	-1.000000	0.000000	-0.500000	1.312500
2	-1.000000	-0.500000	-0.750000	-0.089844
3	-0.750000	-0.500000	-0.625000	0.578369
4	-0.750000	-0.625000	-0.687500	0.232681
5	-0.750000	-0.687500	-0.718750	0.068086
6	-0.750000	-0.718750	-0.734375	-0.011768

$$\boxed{\begin{aligned} [-2,-1]: &\quad x \approx -1.41, \\ [0,2]: &\quad x \approx 1.41, \\ [2,3]: &\quad x \approx 2.73, \\ [-1,0]: &\quad x \approx -0.73. \end{aligned}}$$

9. **a.** Sketch the graphs of $y = e^x - 2$ and $y = \cos(e^x - 2)$.

b. Use the Bisection method to find an approximation to within 10^{-5} to a value in $[0.5, 1.5]$ with $e^x - 2 = \cos(e^x - 2)$.



n	a_n	b_n	x_n	$f(x_n)$
1	0.500000	1.500000	1.000000	-0.034656
2	1.000000	1.500000	1.250000	1.409976
3	1.000000	1.250000	1.125000	0.609080
4	1.000000	1.125000	1.062500	0.266982
5	1.000000	1.062500	1.031250	0.111148
6	1.000000	1.031250	1.015625	0.037003
7	1.000000	1.015625	1.0078125	0.000864
8	1.000000	1.0078125	1.0039063	-0.016973
9	1.0039063	1.0078125	1.0058594	-0.008073

10	1.0058594	1.0078125	1.0068359	-0.003609
11	1.0068359	1.0078125	1.0073242	-0.001374
12	1.0073242	1.0078125	1.0075684	-0.000255
13	1.0075684	1.0078125	1.0076904	0.000305
14	1.0075684	1.0076904	1.0076294	0.0000249
15	1.0075684	1.0076294	1.0075989	-0.000115
16	1.0075989	1.0076294	1.0076141	-0.0000451
17	1.0076141	1.0076294	1.0076218	-0.0000101

$$[1.0076218,\ 1.0076294].$$

$$1.0076294 - 1.0076218 \approx 7.63 \times 10^{-6}.$$

$$\frac{7.63 \times 10^{-6}}{2} \approx 3.81 \times 10^{-6} < 10^{-5}.$$

$$\boxed{x \approx 1.00762}.$$

$$\boxed{\boxed{x = 1.00762}}.$$

$$\boxed{\begin{array}{c} e^x - 2 = \cos(e^x - 2), \\ x \approx 1.00762. \end{array}}$$

s approximately

$$y = e^{1.00762} - 2 \approx 0.7372.$$

approximately at

$$\boxed{(1.00762,\ 0.7372)}.$$

13. Find an approximation to $\sqrt[3]{25}$ correct to within 10^{-4} using the Bisection Algorithm.

The third root of 25 is approximately $p_{14} = 2.92401$, using $[2, 3]$.